ROB-05

TEACHER GUIDE

Robotics · Motors & Motion Control

Build and Wire a Robot Car

Two motors, one driver, one brain. Let's make it roll.

1 At a glance

LENGTH	TYPE	GROUPING	CONCEPTS
90 minutes	Hardware	Pairs, one robot each	H-bridge · differential drive · motor direction

2 Overview

Students build the robot car the rest of the Robotics module uses. They learn why an Arduino pin cannot drive a motor, how an H-bridge reverses one, how PWM sets speed, and how two wheel speeds steer a differential-drive robot. Then they assemble the chassis, wire two motors through an L298N with a separate battery, and write five driving functions that make the robot drive a square.

> The one idea that must land
>
> Arduino pins give signals; the driver gives power. The two grounds must be joined, or nothing the Arduino says means anything to the driver.

3 Learning outcomes

By the end, students can:

1. Assemble a **2WD chassis** so it rolls straight and nothing rubs.
2. Wire two motors through an **L298N driver** and a separate battery, with a shared ground.
3. Explain how an **H-bridge** reverses a motor using four switches.
4. Drive **forward, back, left and right** from code, written as functions.

4 Three depths

Pick one for the class. Everyone does the same lesson; only the build target changes.

Starter

Assemble and drive with ready-made code

Assemble the chassis and wire it with close support, then upload a ready-made square sketch and watch it drive. Focus on why the battery and grounds matter, not on the code.

Core

Wire and code forward, back, left, right as functions

Full lesson: build, wire, make each wheel turn both ways (step 3), then write `drive()` and the five functions and drive a square (step 4).

Advanced

Add speed control and a calibration routine for straight driving

Add a trim constant so the robot drives straight over 2 m, then a soft start that ramps speed over 300 ms. Measure stall current with a multimeter.

5 Before the lesson

Kit, per group of 2

- 2WD robot chassis kit: plate, 2 gear motors, 2 wheels, caster, screws
- L298N motor driver board
- Arduino Uno + USB cable
- 4×AA battery pack with switch, and fresh AA cells
- Male-female jumper wires
- Small screwdriver set

Simulators

- **Circuit Lab**: the L298N, two motors and the battery can be wired and driven on screen; good for checking wiring before power.
- **GearsBot** (online): drive a simulated differential-drive robot to feel left/right steering before the build.

Prep checklist

- [ ] Pre-solder or crimp leads onto the gear motors if the kit motors have bare tabs. Soldering in class needs a teacher present.
- [ ] Fit fresh AA cells, or test packs with a multimeter: under 5.5 V the robot will be sluggish.
- [ ] Check every L298N has its ENA and ENB jumpers; students remove them in step 2.
- [ ] Clear floor space with a taped 1 m square for the final test.
- [ ] Have a box of spare M3 screws and nuts. They will be dropped.

6 Timing plan

Min	Phase	What happens
5	Let's think	Why a pin cannot drive a motor: the current numbers (slide 2).
3	Aims	The four aims and keywords (slide 3).
17	Teach	H-bridge, reading the L298N, PWM, differential drive, real examples (slides 4 to 8).
5	Kit and safety	Kit check and the four safety rules (slide 10).
15	Assemble the chassis	Motors, wheels, caster, battery over the driven wheels (slide 12).
15	Wire it up	Motors to OUT1-4, battery, shared GND, IN and EN pins (slides 11, 13).
10	One wheel turns	Wheels off the desk; each wheel forward and back (slide 14).
12	Four functions	drive(), forward/back/left/right/halt, drive a square (slide 15).
8	Test and wrap up	Make it go straight, self-rating and exit ticket (slides 16, 18, 19).

Total: 90 minutes

7 Step by step

Let's think

5 min · slides 1-2

Say: An Arduino pin gives about 20 mA. A small gear motor wants 150 to 250 mA, and over 1 A when it starts. What happens if we plug a motor straight into a pin?

Say: And to run a motor backwards you must reverse the voltage. Can a pin go negative?

→ Write 20 mA and 1000 mA on the board side by side. Let the ratio do the explaining.

Look for: students who think a bigger delay or a higher number in code will help. The limit is physical, not software.

Aims

3 min · slides 3

Say: By the end of today your robot drives a square and comes back.

→ Read the aims; point out the keywords H-bridge, PWM and common ground.

Teach

17 min · slides 4-8

Say: An H-bridge is four switches round a motor. Close S1 and S4 and current goes one way. Close S2 and S3 and it goes the other way. Never S1 and S3 together: that is a short circuit.

Say: On the L298N, the IN pins decide direction and the EN pins decide speed.

Say: PWM switches the power on and off very fast. The share of time it is on sets the speed. Switching wastes almost nothing; a resistor would waste it as heat.

Say: There is no steering wheel. Two wheels at different speeds make the robot turn.

→ Slide 4: draw the H on the board and have students say which switches close for forward and backward.

→ Slide 6: below about analogWrite(60) cheap motors only buzz. Tell them to start tests at 100.

→ Slide 7: walk through the table with your hands as two wheels.

Look for: the idea that the Arduino 'powers' the motor. Keep saying: the battery powers it, the Arduino only decides.

Kit and safety

5 min · slides 9-10

Say: Wheels off the desk for every first test. Battery in last, out first, whenever you change wiring.

- → Pairs check their kit against the list before anything is unscrewed.
- → Long hair tied back now, before motors run.

Assemble the chassis

15 min · slides 12

Say: Weight goes over the driven wheels, so they grip. Weight over the caster just pushes a sliding lump.

- → Bolt motors in without overtightening; the plastic strips.
- → Push wheels on until flat against the shaft; fit the caster at the opposite end.
- → Spin each wheel by hand: it must turn freely with no rubbing.

Look for: a wheel rubbing the chassis. It will curve the robot however good the code is. Loosen, square, retighten.

Wire it up

15 min · slides 11, 13

Say: Two supplies, one rule: the battery powers the motors, USB powers the Arduino, the grounds are shared, the 5 V rails are not.

- → Follow slide 13 in order: motors to OUT1/OUT2 and OUT3/OUT4; battery + to +12V and – to GND; Arduino GND to driver GND.
- → IN1→7, IN2→8, IN3→9, IN4→11; ENA→5, ENB→6, after removing the two enable jumpers.
- → Check every pair's ground wire before any battery goes in.

Look for: pin 10 left free on purpose. Pins 5 and 6 are PWM for speed; 7, 8, 9, 11 only set direction.

One wheel turns

10 min · slides 14

Say: Wrong direction? Do not change the code. Swap the two motor wires at the driver. Same fix, honest code.

- → Upload the step 3 sketch with the robot on a box, wheels free.
- → Left wheel forward, then swap IN1 and IN2 in code to see it reverse. Repeat for the right wheel with IN3, IN4, ENB.

Four functions

12 min · slides 15

Say: One `drive(left, right)` function does all the work. Forward, back, left, right and halt are each one line that calls it.

- → Students add the step 4 code (with pinMode for all six pins in setup).
- → Drive a square on the floor; tune the 420 ms turn until the robot comes back to its start.

Look for: the `halt()` name. The slide calls it halt because `stop` is easy to confuse; any name is fine if it is used consistently.

Test and wrap up

8 min · slides 16-19

Say: Every real robot has a trim value somewhere. Finding yours is engineering, not cheating.

- → Pairs pick one symptom from slide 16 and fix it, or note it.
- → Self-rate against the proofs; exit ticket: two fixes for drifting left, one mechanical and one in code.

8 Common misconceptions

Students may think	How to help
The Arduino powers the motors.	The battery does. The Arduino sends a small signal telling the driver which way and how fast.
Lower speed means lower voltage.	PWM keeps the voltage and changes how much of the time it is on. That is why it wastes so little.
Turning needs a steering motor.	Differential drive turns by giving the two wheels different speeds or directions.
If both motors get 170 the robot goes straight.	No two motors are identical. Real robots need a trim value, found by testing.
The ground wire is optional if the battery is connected.	Without a shared ground the driver cannot tell HIGH from LOW, so the motors twitch randomly.

9 Troubleshooting

Problem	Fix
Wheels twitch randomly	Join Arduino GND to driver GND.
One wheel does nothing	A loose OUT screw terminal. Tug the wire gently, retighten, retest. Check its ENA/ENB wire too.
Robot curves to one side	Check for rubbing first, then lower the faster motor's speed (for example 170 and 155).
Works on USB, dies on battery	Flat cells, or the pack is too weak. Fit fresh AAs; the L298N itself loses about 2 V.
Arduino resets when motors start	Motor start-up current drags the supply down. Keep the Arduino on USB, motors on the battery, grounds joined.
Motors only buzz	Speed too low: analogWrite below about 60 cannot overcome friction. Start at 100 or more.
Driver gets hot	Disconnect at once. Check nothing shorts the OUT terminals and the motors are not stalled against the chassis.

10 Assessment rubric

	Emerging	Secure	Extending
Building	Chassis assembled with help; a wheel rubs.	Both wheels spin freely; battery over the driven wheels.	Robot rolls straight unpowered; explains why weight placement matters.
Wiring and power	Motors run with help; ground missed at first.	Both motors run both ways; shared ground explained.	Diagnoses a wiring fault from its symptom alone.
Code and steering	Runs the given sketch.	Writes drive() and the five functions; drives a square.	Adds trim or a soft start and shows the improvement with measurements.

11 Answers

Slide 2: Why can't the Arduino drive a motor directly?

> A pin supplies about 20 mA safely; a motor needs 150 to 250 mA running and over 1 A at start. Also a pin cannot reverse the voltage to run the motor backwards.

Slide 4: Which switches close for forward, and for backward?

> Forward: S1 and S4. Backward: S2 and S3. Never S1 with S3 (or S2 with S4): that shorts the battery.

Slide 7: Left wheel forward, right wheel backward. What does the robot do?

> It spins on the spot to the right.

Exit ticket: the robot drifts left. Two fixes?

> Mechanical: check the right wheel is not rubbing, the wheels are on fully and the weight is even. Code: slow the left motor slightly (a trim value), or speed up the right.

12 Safety

- ! Wheels off the desk for every first power-up.
- ! Battery in last, out first, whenever wiring changes.
- ! Tie long hair back and keep loose sleeves away from the wheels: gear motors catch them.
- ! If the driver smells hot or the batteries get warm, disconnect at once and tell a teacher.
- ! Alkaline AA cells only. Never mix old and new cells, and never short a battery pack's leads.